SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA14 **COURSE NAME: Compiler Design**

COURSE OUTCOME COVERED

CO1: Apply lexical analysis for token specification and recognition using LEX tool. *(BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**

Mode: In Class **Hours : 60 mts**

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I S.MAHALAKSHMI(192524372) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
S.MAHALAKSHMI
Signature of the student:

Q. No	Problem Understanding (20)	Method Selection (20)	Solution Process (25)	Accuracy of Calculation (20)	Interpretation of Results (15)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	**/ 20**	**/ 25**	**/ 20**	**/ 15**	**/ 100**

Annexure A

ANALYTICAL PROBLEM SOLVING

1	Trace the processing of the input expression P = (a * b) + (b * c) * 2 through all phases of a compiler. a) Identify and list all tokens generated during lexical analysis. b) Construct the syntax tree during the parsing phase. c) Generate the intermediate code for the given expression. d) Apply suitable optimizations to the intermediate code. e) Explain the significance of the final target code prod
2	For each of the following Regular Expressions, construct the language. Write any FIVE valid strings for each Regular Expression. a)$(\varepsilon+aa^*)^*$ b)$(a^*+b^*)^*\ abb$ c)$(a^*b^*)^*\ aba\ (a^*+b^*)^*$ d)$(a+ab)^*$ e) $(aba)^*+(a^*b^*)^*$

3	i)Find the number of tokens in the following C statement is
	main( )
	{
	int a=10;
	char b =”abc”;
	in t c = 30;
	ch ar d = “xyz”;
	in /* comment */ t m = 40.5;
	}
	ii)Identify the lexical errors in the following C statement is
	void main()
	{
	int x=10, y=20;
	char * a;
	a= &x;
	x= 1xab;
	}

4	**Find a regular expression corresponding to each of the following subsets of** $\{0,1\}^*$**:** i) The language of all strings that have an even number of 0’s. ii) The language of all strings that contain at least one 1. iii) The language of all strings in which both the number of 0’s and the number of 1’s are odd. iv) The language of all strings that start with 1 and end with 0. v) The language of all strings that do not contain consecutive 1’s.

5	**Consider a lexical analyzer for a simplified programming language. The language consists of the following tokens:** 1. Keywords: if, else, while, return, int, float 2. Operators: +, -, *, /, =, ==, <, > 3. Separators: (,), {, }, ;, , 4. Identifiers: A sequence of letters followed by a sequence of digits, e.g., var123 5. Integer Literals: A sequence of digits, e.g., 12345 6. Floating-point Literals: A sequence of digits followed by a dot and another sequence of digits, e.g., 123.45 **The lexical analyzer needs to tokenize the following input string:** int var123 = 12345; float var456 = 123.45; if (var123 == var456) { return; } **Questions:** a) How many tokens are there in the given input string? b) Provide a list of tokens categorized by their types (keywords, operators, separators, identifiers, integer literals, floating-point literals). c) Assuming a finite state machine (FSM) is used for lexical analysis, estimate the number of state transitions required to tokenize the given input string. Provide your reasoning based on the transitions needed for each token type.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method

Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

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Annexure A

ANALYTICAL PROBLEM SOLVING

89

Code of Conduct: (192524372)

I S. Hahalakshmi *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
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S. Hahalakshmi
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3	/ 4	/ 4	/ 4	/ 4	/ 4	
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5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 20	/ 25	/ 20	/ 15	/ 100

CO-1 Assesment tool - 1 CSA1410 - Compiler Design	Name:- S. Mahalakshmi Reg no:- 192524372 Department:- B.tech AI DS

Analytical Problem Solving

1) Trace the processing of input expression $P=(a*b)+(b*c)*2$ through all phase

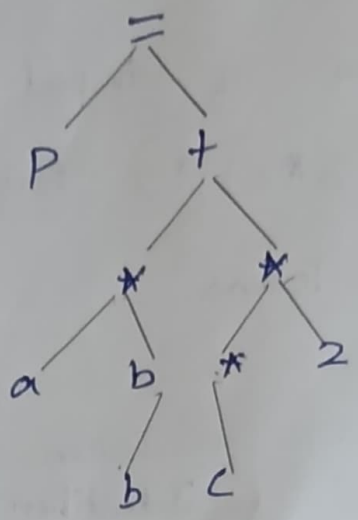
a) Lexical Analysis - Tokens

S.no	Lexeme	Token
1)	P	Identifier
2)	=	Assignment operator
3)	(	Left parenthesis
4)	a	Identifier
5)	*	Multiplication operator
6)	b	Identifier
7)	)	right parenthesis
8)	+	Addition operator
9)	(	Left parenthesis
10)	b	Identifier
11)	*	multiplication operator
12)	c	Identifier
13)	)	Right parenthesis
14)	*	Multiplication operator
15)	2	Number
16)	;	Terminator

$$id = (id * id) + (id * id) * 2$$

b) Syntax Tree

$$P = (a * b) + (b * c) * 2$$



c) Intermediate code

$$t_1 = a * b$$
$$t_2 = b * c$$
$$t_3 = t_2 * 2$$
$$t_4 = t_1 + t_3$$
$$P = t_4$$

d) Code optimization

$$t_1 = a * b$$
$$t_2 = b * c$$
$$t_3 = t_2 << 1$$
$$t_4 = t_1 + t_3$$
$$P = t_4$$

e) Significance of final target code

```
MOV R1, a
MUL R1, b
MOV R2, b
```

```
MUL R2, C
SHL R2, 1
ADD R1, R2
MOV P, R1
```

2) For each of the following regular expression construct the language. Write any five valid string for each regular expression.

a) aa

$L = \{ \epsilon, aa, aaaa, aa\,aa\,aa \}$

five valid strings.

aa
aaaa
aa aa aa
aa aa aaaa
aa aa aa aaaa

b) $(b^*)^* abb$

$(b^*)^* = b^*$

$b^* abb$

five valid strings

abb
babb
bbabb
bbbabb
bbbbabb

c) $(a^*b^*)^* aba (a^* + b^*)^*$

five valid string

aba
aaba
abaa
baba
ababa

4 d) $(ab)^*$

$L = \{\epsilon, ab, abab, ababab \ldots\}$

five valid strings

ab
abab
ababab
abababab
ababababab

e) $(aba)^* + (a^*b^*)^*$

five valid strings

a
b
ab
ba
aab

3) Find the numbers of tokens in the following c statement.

i) Number of tokens.

Main, (,), {, int, a, =, 10, char, b, =,
"abc", ;, int, c, =, 30, char, d, =, "xyz",
;, int, m, =, 40.5, ;, }

The total number of tokens are 30

ii) Identify the lexical errors.

```
Void main( )
{
    int x=10, y=20;
    Char *a;
    a = z x;
```

```
x = 1xab;
}
```

The lexical error :- 1xab

It is invalid because identified cannot starts with digit.

4) Find the regular expression corresponding to each the following subset of {1}:

$$\Sigma = \{0, 1\}$$

i) String having an even number of 0's

$$1^*(01^*01^*)^*$$

ii) strings containing at least one

$$0^*1(0+1)^*$$

iii) The language of all strings in which both the numbers of 0's and the numbers of 1's are odd.

$$(00+11)^*(01+10)(00+11)^*[(01+10)(00+11)^*(01+10)(00+11)^*]^*$$

iv) The language of all strings that start with 1 and end with 0

$$1(0+1)^*0$$

eg:- 10, 100, 1010

v) The language of all strings that do not contain consecutive 1's

$$(0+10)^*(\epsilon+1)$$

5) Consider a lexical Analysis of simplified Performing language.

a) How many tokens are there.

int, var123, =, 12345, ;, float,
var456, =, 12345, i, if, var123, ==, var456,
{, return, ;, }

The total number of tokens are 20

b) Categories the tokens

1) Keywords

int, float, if, return

4 - Keywords

2) Operators

=, =, ==

3 - operators

3) Separators

;, ;, (,), {, ;, }

7 - separators

4) Identifiers

var123, var456, var123, var456

4 - identifiers

5) Integer literals

12345

1 - Integer literals

6) Floating - Point literals

123.45

1 - floating - Point literals

so totally 20 tokens

c) FSM state transition

For each token FSM is based on ch

Token	No. of transition
int	3
var 123	6
=	1
12345	5
;	1
float	5
var456	6
=	1
123.45	6
;	1
if	2
(	1
var123	6
= =	2

var456	6
)	4
{	1
return	6
;	1
}	1

Totally 62 transitions.